Philosophy of Space and Time

Lecture 6: *The Bucket, the Globes, and the Leibniz-Clarke Debate*

Chris Smeenk

Department of Philosophy, UCLA · April 16, 2026

Today's Plan

1. Recap: Newton's Framework
2. The Effects Argument
3. The Bucket Experiment
4. The Globes Thought Experiment
5. The Leibniz-Clarke Debate
6. Assessment and Preview
7. Geroch on Spacetime

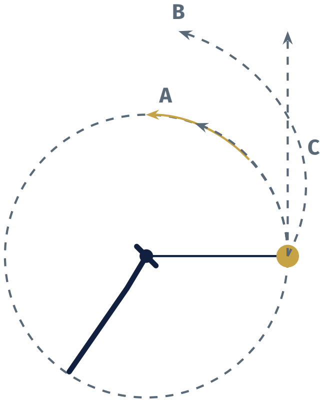
Where We Left Off

Tuesday's lecture traced the central tension between **relativity** and **inertia**:

- Galileo: uniform motion is undetectable $\rightarrow$ motion is relative
- But the law of inertia privileges straight-line uniform motion $\rightarrow$ requires a standard of "straight" and "uniform" that goes beyond relations among bodies
- In *De Gravitatione*, Newton showed that Descartes' relational definition incompatible with Descartes' own laws
- In the Scholium, Newton defended absolute space "by properties, causes, and effects" — we have not yet discussed *effects* argument

Today: the effects argument, then Leibniz's relationalist criticism

Recap: a Stone in a Sling

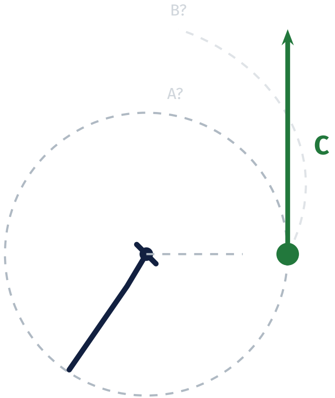


A stone whirls in a sling. At the marked point, the cord is released.

??*What path does the stone follow after release?*

A. Continues in a circle
B. Flies off along a curved (spiral) path
C. Continues in a straight line (tangent to the circle)

A Stone in a Sling

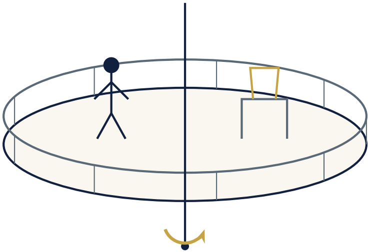


Answer: C — a straight line.

The sling provides the force that continually pulls the stone away from straight-line motion. Once released:

- No net force acts (ignoring gravity and etc.)
- By the **law of inertia**, the stone continues in a straight line at uniform speed
- Along the tangent at the release point

The Rotating Laboratory



Constantly rotating platform

Imagine a scientist inside a laboratory that is mounted on a constantly **rotating** platform.

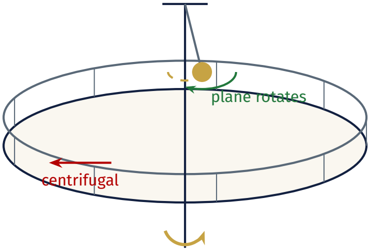
?? *Can the scientist detect that the laboratory is rotating, without looking outside?*

A. Yes

B. No

If **A**: how?

The Rotating Laboratory



Foucault pendulum detects rotation

Answer: A — Yes!

Rotation is *not* uniform straight-line motion. The principle of relativity says only **inertial** (non-accelerating) motion is undetectable.

How to detect it:

- Objects slide outward (centrifugal effects)
- A **Foucault pendulum**: the plane of swinging rotates relative to the floor

An Irony: Galileo's Ship Revisited

Foucault's pendulum, Panthéon, Paris

Galileo: the motion of the Earth is undetectable from "onboard," based on the **principle of relativity**.

But the Earth is *rotating*, and rotation is *not* inertial motion.

In 1851, **Foucault** demonstrated this: a pendulum in the Panthéon in Paris slowly changed its plane of oscillation over the day—revealing the Earth's rotation.

The Scholium, §XII: Effects of True Motion

Completing the Scholium's Argument

Recall the structure of the Scholium:

§§I–VI: Distinctions between absolute and relative space, time, motion ✓

§§VIII–X: **Properties** of absolute motion ✓

§XI: **Causes**: force is necessary and sufficient for changing true motion ✓

§XII: **Effects**: centrifugal effects track true rotation, not relative rotation ← **today**

§XIV: The epistemological problem ✓

The "effects" argument is Newton's most famous and most powerful, and it depends on the Second Law...

Newton's Second Law

Law II

A change in motion is proportional to the motive force impressed and takes place along the straight line in which that force is impressed.

(Modern formulation: $\mathbf{F} = m\mathbf{a}$.)

- The First Law tells us what happens with *no* forces: uniform straight-line motion
- The Second Law tells us what happens *with* forces: acceleration proportional to force
- **Acceleration** is a change in the state of motion—and it has observable consequences

Newton's strategy: use these observable consequences to argue that true (absolute) motion is distinct from relative motion.

Warm-Up: Detecting Acceleration

Before the bucket, consider simpler cases:

The Accelerometer

- A ball on a string hanging in a car
- Car accelerates $\rightarrow$ ball swings backward
- The deflection indicates a force
- You can detect acceleration *without looking outside*

The Coffee Cup

- Stir coffee in a cup, then stop the spoon
- Coffee keeps rotating—surface is concave
- The concavity indicates centrifugal effects
- Rotation has observable consequences

?? *In both cases, the observable effect (deflection, concavity) tracks true acceleration, not mere relative motion. This is exactly the Scholium's point in §XII: the effects of motion distinguish absolute from relative.*

The Bucket Experiment

Setup (Scholium, §XII)

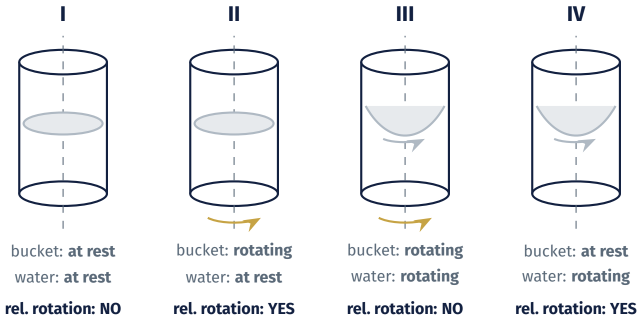
Newton describes a bucket of water suspended by a twisted cord.

- Release the cord: it unwinds, spinning the bucket
- Friction gradually transfers the rotation to the water
- Watch the surface of the water

We can identify **four stages**:

Stage 1	**Stage 2**	**Stage 3**	**Stage 4**
Before release	Just released	Spinning together	Bucket stopped

The Four Stages



(Stage IV due to Reichenbach: stop the bucket suddenly while water still spins.)

The Argument

Compare the stages:

- Stages 2 and 4: water and bucket in relative motion, but water is flat (stage 2) and concave (stage 4)
- Stages 1 and 3: no relative motion between water and bucket, but water is flat (stage 1) and concave (stage 3)

Newton's Conclusion

The shape of the water surface does not correlate with the *relative* motion of water and bucket. It correlates with the *absolute* rotation of the water.

Therefore: relative motion cannot account for the observable effects. There must be a fact about **absolute rotation**—rotation with respect to absolute space.

What Exactly Does the Bucket Show?

?? *The bucket experiment shows that the concavity of the water surface is not explained by the water's motion relative to the bucket. What follows from this?*

Possible responses:

1. **Newton**: Requires absolutes space, rotation is defined with respect to it
2. **Mach** (later): No need for absolute space — the relevant relative motion is with respect to the *distant stars*, not the bucket
3. **Modern**: The experiment shows a need for *some* non-relational structure, but not the full structure of absolute space

The Globes

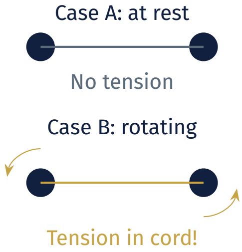
Newton's Two Globes

Imagine two globes connected by a cord, alone in an otherwise *empty* universe.

- Case A: the globes are at rest
- Case B: the globes are rotating about their common center

In case B, there will be **tension** in the cord (centrifugal effect). In case A, there will be none.

But there is *no other body* to be in relative motion with respect to!



Why the Globes Matter

- The bucket experiment can be challenged: maybe the relevant relative motion is with respect to the *lab*, or the *earth*, or the *stars*
- The globes eliminate all of these: there are *no other bodies*
- Yet there is still a physical difference (tension vs. no tension)

Newton's Conclusion

Even in an otherwise empty universe, there is a fact about whether a system is rotating. This fact cannot be reduced to relative motion. Therefore rotation is **absolute**.

?? *Maudlin emphasizes: the relationist has "an insurmountable task overcoming the globes example." Why is this harder than the bucket?*

Mach's Response

Ernst Mach (1883) challenged Newton's thought experiment:

- We have no experience of an *empty* universe—the thought experiment is too far from reality to be a reliable guide
- For *actual* rotation (the bucket, the earth's bulge, hurricanes), the relevant relative motion is with respect to the **distant stars**
- Mach's conjecture: if the stars were somehow rotating around the globes, the cord *would* develop tension

?? *Mach's response is suggestive but not a worked-out theory. Can we actually construct a physics where rotation is fully relational? (This remains an open question.)*

“The Situation Is Not Entirely Desperate” (§XIV)

Newton continues:

> *“Now if some distant bodies were set in that space and maintained given positions with respect to one another, as the fixed stars do . . . it could not, of course, be known from the relative change of position of the balls among the bodies whether the motion was to be attributed to the bodies or to the balls. But if the cord was examined and its tension was discovered to be the very one which the motion of the balls required, it would be valid to conclude that the motion belonged to the balls and that the bodies were at rest.”*
>
> — Newton, *Principia*, Scholium

The key move: even though absolute space is imperceptible, the **effects** of absolute motion (forces, tensions) give us empirical access to the true motions.

Newton's Empirical Strategy

Newton closes the Scholium by promising that the true motions *can* be found:

> *"But in what follows, a fuller explanation will be given of how to determine true motions from their causes, effects, and apparent differences . . . For this was the purpose for which I composed the following treatise."*
>
> — Newton, *Principia*, Scholium

How does this work in practice?

Newton’s Empirical Strategy

1. All observations are made with respect to some **relative space** (the lab, the Earth, etc.)
2. If that relative space is itself accelerating or rotating, observed accelerations will not match the true forces among bodies—*unexplained* accelerations appear
3. Such discrepancies signal that the relative space is not suitable; one corrects by choosing a better reference
4. The *Principia*’s Book III carries this out for the solar system: from observed planetary motions, Newton infers the force of gravity, determines the true motions, and identifies where our initial reference frame went wrong

This iterative, self-correcting method is the empirical payoff of Newton’s framework.

Break

The Leibniz-Clarke Debate

Historical Background

- 1715–1716: correspondence between Leibniz and Samuel Clarke (Newton's proxy)
- Leibniz attacked Newtonian physics on philosophical and theological grounds
- Clarke defended Newton, with Newton himself likely involved behind the scenes
- Leibniz died in 1716; Clarke published the correspondence

The debate is one of the great set pieces in the history of natural philosophy.

Leibniz's Two Principles

Leibniz's arguments against absolute space rest on two metaphysical principles:

Principle of Sufficient Reason (PSR)

Nothing happens without a *sufficient reason* why it should be so rather than otherwise.

Identity of Indiscernibles (PII)

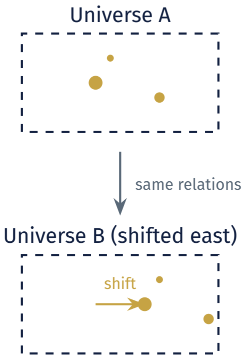
There cannot be two things that are exactly alike in every respect—if they are indiscernible, they are identical.

Leibniz uses these to argue that absolute space leads to absurdities.

The Static Shift Argument

Leibniz's argument (Third Paper, §5):

1. Suppose absolute space exists
2. God could have created the universe *shifted 3 miles to the east*
3. The two universes would be identical in all relational properties
4. But they would differ in which points of absolute space are occupied
5. By the PII, they are not really distinct
6. By the PSR, God would have no reason to choose one over the other
7. Contradiction → absolute space does not exist



The Kinematic Shift

Clarke (and Newton) recognized a further version:

- Instead of shifting everything in space, set everything in **uniform motion**
- Recall Galileo's ship from Tuesday: no experiment below decks can detect uniform motion
- Newton himself proves this as Corollary V: physics is the same in any uniformly moving frame
- So even Newton admits that *absolute velocity* is undetectable

?? *The kinematic shift is Leibniz's version of the very principle of Galilean relativity that Newton endorses. If absolute velocity is undetectable, why should we include it?*

Leibniz's Three Shift Arguments

Static shift	Kinematic shift	Dynamical shift
Relocate everything in space Against: absolute *position*	Set everything in uniform motion Against: absolute *velocity*	Accelerate everything uniformly Against: absolute *acceleration*
Newton vulnerable	Newton vulnerable	**Leibniz vulnerable**

The bucket and globes show that absolute *acceleration* has observable effects.
So the dynamical shift fails against Newton.

But the first two succeed: Newton postulates more structure than his physics needs.

How Leibniz Fails to Meet Newton's Challenge

- Leibniz's shift arguments: isolate excess structure in Newton's account
- But Leibniz never provides a *positive* account of how to explain the bucket and globes relationally
- In his Fifth Paper (§53), Leibniz seems to concede that there is a distinction between true and relative motion—but never explains how this fits with relationalism

> *"I grant there is a difference between an absolute true motion of a body, and a mere relative change of its situation with respect to another body."*
>
> — Leibniz, Fifth Paper, §53

Maudlin: Leibniz "never meets the challenge that Newton lays down."

Responses to Leibniz's Arguments

How might Newton (or a Newtonian) respond?

1. **Deny the PII**: What's wrong with distinct situations that we can't tell apart? (Allow physics to introduce theoretical structure that outruns observation.)
2. **Deny the PSR**: God might make a choice without having a sufficient reason to prefer one option over another.
3. **Accept the lesson partially**: The static and kinematic shifts show that Newton postulates too much structure; but the dynamical structure (acceleration) is needed.

Option (3) leads to the idea of **Galilean spacetime**—which we will explore next week.

Taking Stock: The Relativity/Inertia Tension Revisited

Where does the debate stand?

Newton's Strengths

- The bucket and globes show that *inertia* demands non-relational structure
- His physics works: the *Principia* is spectacularly successful
- Forces and accelerations are real and observable

Newton's Vulnerabilities

- *Relativity* (Galileo's ship, Corollary V) shows absolute velocity is unobservable
- The shift arguments reveal genuine redundancy in absolute position and velocity
- Absolute space is imperceptible

The question for next week: can we strip away the excess structure and keep only what inertia requires?

Summary

1. The Scholium's "effects" argument (§XII): the **bucket experiment** shows that the shape of the water correlates with absolute rotation, not relative motion
2. The **globes**: even in an empty universe, rotation has observable effects, this poses a challenge for the relationist
3. **Geroch's spacetime framework**: starting from events and asking what structure must be added clarifies what Newton assumes (and what might be removable)
4. The **Leibniz-Clarke debate**: Leibniz's shift arguments show that Newton postulates too much structure; but Leibniz never explains how to manage without the structure needed for inertia
5. The **relativity/inertia tension** remains: Galilean relativity undermines absolute velocity, but the bucket and globes defend absolute acceleration

For Next Time

- **Read:** Maudlin, Ch. 3 (Eliminating Unobservable Structure; Galilean Spacetime).
- **Think about:**
  - Leibniz's shift arguments show Newton has too much structure. The bucket and globes show we need *some* non-relational structure. Can we find the right amount?
  - Can we have absolute *acceleration* without absolute *velocity*?

Tuesday: **Eliminating Unobservable Structure**

- Galilean spacetime: keeping only the structure physics needs
- A partial resolution of the Newton–Leibniz debate
- What remains of the relativity/inertia tension?

Spacetime:
Geroch’s Framework

From Space and Time to Spacetime

So far we have been talking about *space* and *time* as separate entities. Geroch (Chs. 1–2) introduces a different starting point:

The Basic Concept

Spacetime is the collection of all *events*—all the things that happen at a particular place and time.

- An **event** is maximally specific: “this flash of light, here, now”
- Events are the raw data of physics; everything else is structure we impose on them
- Both “where” and “when” are folded into a single four-dimensional entity

This shift in perspective turns out to be enormously powerful.

Why Start with Events?

Starting with events is neutral between Newton and his critics:

- Newton's picture: spacetime $=$ absolute space $\times$ absolute time, with a fixed way of slicing spacetime into "space at an instant"
- A relationist might accept the collection of events but deny that they are embedded in an independently existing container
- (Einstein will later show that spacetime cannot be neatly separated into space and time – so we really need spacetime)

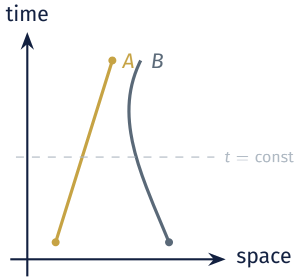
?? *By starting with events and asking "what structure must we add?" Geroch lets us compare different theories of space and time on equal footing.*

Observers and Worldlines

An **observer** traces out a continuous path through spacetime—a **worldline**.

- Each point on the worldline is an event in the observer's history
- The observer carries a clock that records the elapsed time along their worldline
- Different observers may disagree about which events are "simultaneous"

In Newtonian physics, there is a *fact* about simultaneity (absolute time). In relativistic physics, there is not.



What Structure Does Spacetime Need?

Geroch's framework lets us pose the central question precisely:

- Start with the bare collection of events (a set with topology)
- Add structure as needed:
  - A notion of **simultaneity** (which events happen "at the same time")?
  - A **temporal metric** (how much time elapses between events)?
  - A **spatial metric** (how far apart are simultaneous events)?
  - A notion of **straight worldline** (which observers are inertial)?

Newton's spacetime has *all* of these, plus more (absolute rest, absolute velocity). The question raised by the Leibniz-Clarke debate: can we get by with less?

Newtonian Spacetime in Geroch's Terms

In Geroch's framework, Newton's spacetime has the structure:

Absolute **simultaneity**: a fixed slicing into "instants"

Temporal **metric**: how much time elapses between instants

Spatial **metric**: Euclidean distance within each instant

Persistence of spatial points through time ("same place")

Absolute **velocity**: a fact about which worldlines are "at rest"

Overflow: Coordinates vs. Geometry (Maudlin, Ch. 2)

An important methodological distinction:

Key Distinction

Coordinates are labels for points of space. They are not themselves spatial structure.

- A coordinate system assigns numbers (x, y, z) to each point
- Different coordinate systems assign different numbers to the same point
- The *geometry* of space is what remains invariant under changes of coordinates
- Newton's *absolute space* is a claim about geometry, not coordinates
- Many confusions arise from conflating coordinate-dependent descriptions with genuine physical structure

Overflow: The Earth's Bulge

A real-world instance of Newton's argument:

- The earth is not a perfect sphere—it bulges at the equator
- This bulge is due to the centrifugal effect of the earth's rotation
- The bulge cannot be explained by the earth's rotation *relative to the sun or stars alone*—it is an effect of absolute (or at least non-relational) rotation
- Newton predicted the size of the bulge from his theory; it was later confirmed by geodetic measurements

Overflow: Corollary VI and Accelerated Frames

> *"If bodies are moving in any way whatsoever with respect to one another and are urged by equal accelerative forces along parallel lines, they will all continue to move with respect to one another in the same way as they would if they were not acted on by those forces."*
>
> — Newton, *Principia*, Corollary VI

This is more general than Corollary V: even certain *accelerated* frames are equivalent—if all bodies are equally accelerated.

This anticipates Einstein's equivalence principle and the move to general relativity.

Overflow: Three Relationalist Theses (Earman)

Following Earman (1989), we can distinguish three versions of relationalism:

- **R1**: All motion is the relative motion of bodies; spacetime has no structure supporting absolute quantities of motion.
- **R2**: Spatiotemporal relations among bodies are *direct*, not mediated by an underlying spatial substratum.
- **R3**: There are no irreducible monadic spatiotemporal properties (like "is located at point *p*").

Newton's arguments attack R1 directly. But R2 and R3 can be maintained even if R1 fails—there is logical space for intermediate positions.

Overflow: Sklar's Maneuver

Lawrence Sklar proposed a way to block the inference from ¬R1 to ¬R2:

- Accept that there is absolute acceleration
- But treat acceleration as a **primitive property** of bodies, not defined via motion through absolute space
- This avoids postulating absolute space while acknowledging that acceleration is non-relational

?? *Is this a genuine middle ground, or does it just relabel the problem?*

Questions?

chris.smeenk@gmail.com